

Yafira Martinez

Design Engineer / Creative Technologist

interactive systems, interfaces, and experimental technologies

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EXPERIENCE

Teaching and Academic Support

Graduate Teaching Assistant — Intro to Wearables, NYU (IMA) Jan 2026 – May 2026

Adjunct Instructor: Daniel Ryan Johnston

- Mentored students in wearable and e-textile design, focusing on body-centered interaction, gesture, and non-verbal communication.
- Assisted students with hands-on projects with soft circuits and Adafruit Gemma M0, using Arduino code with light and motion sensors to prototype expressive wearables.

Graduate Teaching Assistant — Creative Computing, NYU (IMA) Sep 2025 – Dec 2025

Professor: Daniel Shiffman

- Supported students in creative coding through office hours, workshops, and assignment feedback.

Research Assistant — Computational Text, NYU (ITP/IMA) May 2025 – Aug 2025

Professor: Daniel Shiffman

- Assisted in developing the Computational Text course syllabus by identifying tools, artist projects, and frameworks in procedural text, data ethics, and machine learning.
- Researched and curated readings and case studies bridging procedural text, data ethics, and machine learning for creative coding contexts.

Research & Open Source

Hardware Research Developer — NYU Ability Lab Oct 2025 – May 2026

- Researched accessible fashion technologies for the Making Fashion Accessible project using knitting machines and AYAB to support inclusive, open-source garment production.
- Contributed to adaptable textile workflows centered on accessibility and human-centered design.

Machine Learning Researcher & Web Contributor — ml5.js Jun 2025 – May 2026

- Developed ml5.transcribe() and ml5.embeddings(), expanding access to browser-based transformer models (Whisper, Moonshine) for creative coding.
- Refined interaction design, inference performance, and web tooling to support real-time creative ML and contributor usability.

Development & Community

Lead Camp Counselor — ITP Camp, New York University Summers 2025 – 2026

- Supporting a 150+ person creative technology community through workshops, project mentorship, and hands-on troubleshooting across hardware, software, and fabrication during a month-long intensive.
- Mentoring first-year counselors across day-to-day camp operations.

Lead Web Developer — Equitable Syllabus Project, NYU (ITP/IMA) Nov 2024 – May 2026

- Designed and implemented accessible, user-centered interfaces using React.js and Figma.
- Integrated and migrated project data while maintaining centralized contribution tracking (Google Sheets) to improve usability and sustainability.

Web Developer & Designer — Useless Machines Archive Mar 2025 – Aug 2025

Professor: Blair Simmons

- Designed and built a front-end web archive using the Are.na API, focusing on client-side data fetching, content organization, and responsive interface design.

DESIGN

UX/UI Design, Interaction Design, Responsive Design, Prototyping, Wireframing, Design Systems, Digital Accessibility, Information Architecture, User Research, Figma, Adobe Suite

SOFTWARE

JavaScript, HTML/CSS, React, Next.js, Node.js, APIs/REST APIs, Web Serial, p5.js, ml5.js, Python, CircuitPython, Creative Coding, Git

HARDWARE

Physical Computing, E-Textiles, Soft Circuits, Wearables, Soldering, Fabrication, Microcontrollers, Arduino IDE, Raspberry Pi, Adafruit, Fritzing, Laser Cutter, Knitting Machine

CERTIFICATIONS

Artificial Intelligence,

Samsung Innovation Campus

November 2022 – May 2023

Human Computer Interaction,

Interaction Design Foundation

February 2022

Google UX Design, Coursera

July 2021 – December 2021

EDUCATION

Master of Professional Studies, Interactive Telecommunications Program (ITP)

New York University –

Tisch School of Arts

May 2026

Bachelor of Science,

Computer Science

City University of New York -

Lehman College

December 2020

SELECTED WORK & FEATURES

PetalByte — Bring Your Own Data (BYOD), NY Tech Week, NYC

June 2026

- Exhibited a portable generative text device treating the USB drive as both archive and interface. Personal corpora and poetic fragments generated via Markov chain live on a flower-shaped laser-cut acrylic enclosure, browsable through a custom pygame interface on a Raspberry Pi touchscreen display.

The Soft Computer — Thesis, NYU's Interactive Telecommunications Program (ITP)

May 2026

- Textile-based computing interface combining machine-knit wool, capacitive fabric inputs, haptics, and a flexible e-ink display. Raspberry Pi 5 drives Markov chain text generation across four corpus-mapped channels. Exhibited at Open Hardware Summit, Berlin.

Soft Circuit Library — Electronics Faire (Theme: Repair), Temple University Libraries

April 2026

- Exhibited a collection of handmade textile circuit boards organized as a living archive, each designed to be handled, studied, and repaired by touch. Visible repair notes and patched connections treat maintenance as part of the circuit's ongoing life.

Guardians of the Groceries (G.O.T.G.), — PlayfulNYC Hackathon 2026, Parsons School of Design

April 2026

- Built the hardware and software layer of a handheld ESP32-based device that pairs with a live web registry to document NYC bodega cats. It captures a photo, uploads it to Supabase via WiFi, and logs each cat's alter ego, hidden powers, and cuteness rating. **Awarded** ☆ **Best of Software** ☆

Open Hardware Summit Fellow — Open Source Hardware Association, Berlin, Germany

Feb 2026 - present

- Selected fellow exhibiting The Soft Computer and other experimental open-source hardware projects at the Open Hardware Summit, Berlin.

Pocket Oracle of Time — Horological Society of New York (HSNY)

Feb 2026 - April 2026

- Exhibited a poetic timekeeping object exploring ritual, slowness, and material interaction with time through embedded electronics and tactile design.

MIT Hard Mode Hackathon, AI x Hardware 2026 — MIT Media Lab

Mar 2026

- Built *enoki*, a focus companion for communal accountability — a React Native app using Mentra smart glasses and Claude Vision AI paired with a desktop companion device using Raspberry Pi, Sreed sensors, and motors, both designed to work together and visualize focus as a mushroom that grows when on task and wilts when distracted.

Moon Pocket — Pedestal Playtime: Art Toy & Tech Toy Open Mic, NYC Resistor

Jan 2026

- Showcased a connected device built with an Arduino Nano 33 IoT that fetches real-time moon and tide data from public web APIs and renders it on a round TFT display through touch-based interaction.

Tendertronics: Analog Circuit Performance — NYU ITP's NIME Show, Littlefield

Dec 2025

- Performed an analog instrument exploring tactile, minimal interfaces and embodied sound interaction through handcrafted circuitry.

Lightning Talk Speaker — Exploring Creativity in the Age of AI, NYU Global Show & Tell

Oct 2025

- Presented *Open Tools, Soft Interfaces*, discussing contributions to ml5.js and open-source machine learning, and framing creativity in AI through openness, design as dialogue, and soft, human-centered interfaces.

Static Chic — Volt Couture: Wearables + E-textiles Open Mic, NYC Resistor

May 2025

- Presented *Static Chic*, a 3D-printed inductive handbag that wirelessly lights up embedded LEDs. Part of a wearables and e-textiles open mic curated by Kari Love and Liza Stark.

MIT Reality Hack 2025 — Massachusetts Institute of Technology

Jan 2025

- Implemented sound and haptic feedback for *HangarPro VR*, a virtual reality training simulation using Cognitive3D to enhance pre-flight safety through visual interaction analysis for aerospace exploration.